

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0703

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes'

1. Industry Problem

A 20-person startup requires a reliable and easily manageable computer network to connect employee workstations, printers, VoIP phones, laptops, and other network devices. The network should provide good performance, easy troubleshooting, fault isolation, and the ability to expand as the company grows.

2. Proposed Solution

A star topology is proposed for the startup. In this topology, all network devices are connected to a central managed switch. A 24-port switch can be used to connect the 20 employee computers while keeping additional ports available for printers, VoIP phones, wireless access points, and future devices.

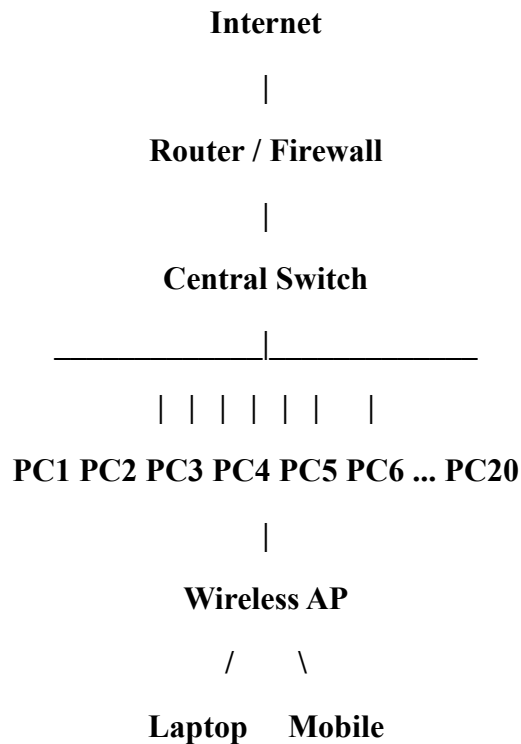
The switch is connected to a router/firewall, which provides Internet connectivity, DHCP, and basic network security. A wireless access point can also be connected to the switch to provide Wi-Fi access for laptops and mobile devices.

3. Network Components

- 1 × Managed 24-port switch
- 1 × Router/firewall

- 20 × PCs/workstations
- 1 × Wireless access point
- Cat6 UTP Ethernet cables
- Internet connection

4. Network Design



The central switch acts as the main connection point for all devices. Each workstation has a dedicated connection to the switch.

5. IP Addressing Scheme

The network can use the private IP range 192.168.1.0/24.

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.1.1	255.255.255.0	—
PC1	192.168.1.2	255.255.255.0	192.168.1.1
PC2	192.168.1.3	255.255.255.0	192.168.1.1
PC3	192.168.1.4	255.255.255.0	192.168.1.1
PC4	192.168.1.5	255.255.255.0	192.168.1.1
...	...	...	...
PC20	192.168.1.21	255.255.255.0	192.168.1.1

6. Implementation in Cisco Packet Tracer

The network can be implemented using Cisco Packet Tracer.

Steps:

1. Open Cisco Packet Tracer.
2. Place one router and one 24-port switch.
3. Add 20 PCs to represent the employees' workstations.
4. Connect all PCs to the central switch using Copper Straight-Through cables.
5. Connect the switch to the router.
6. Configure the router interface with IP address 192.168.1.1.
7. Configure IP addresses for the PCs from 192.168.1.2 to 192.168.1.21.
8. Set 255.255.255.0 as the subnet mask.
9. Set 192.168.1.1 as the default gateway for the PCs.
10. Verify connectivity using the ping command.

7. Testing and Verification

From PC1, open:

Desktop → Command Prompt

Enter:

```
ping 192.168.1.3
```

A successful response indicates that PC1 can communicate with PC2 through the central switch.

Example:

```
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
```

8. Problem-Solving Benefits

The proposed star topology solves several practical industry problems:

- Easy troubleshooting: Each device has an individual connection to the switch.

- Fault isolation: Failure of one cable or PC does not affect the other devices.
- Scalability: Additional devices can be connected using the available switch ports.
- Centralized management: The network administrator can manage connected devices through the central switch.
- Good performance: Dedicated Ethernet connections provide reliable communication.
- Wireless support: An access point allows laptops and mobile devices to connect wirelessly.
- Future expansion: Spare switch ports allow the startup to add more employees and devices.

9. Expected Result

The star topology provides the startup with a reliable, scalable, and easy-to-manage network. All 20 workstations can communicate through the central switch, while the router provides access to external networks. Successful ping responses confirm that the network is functioning correctly.

Conclusion

A star topology is an appropriate solution for a 20-person startup because it provides centralized management, simple troubleshooting, fault isolation, good performance, and easy future expansion. The design can be successfully implemented and tested using Cisco Packet Tracer.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Industry Problem

A bank requires near-zero downtime because network failure can interrupt ATM services, online banking, and financial transactions.

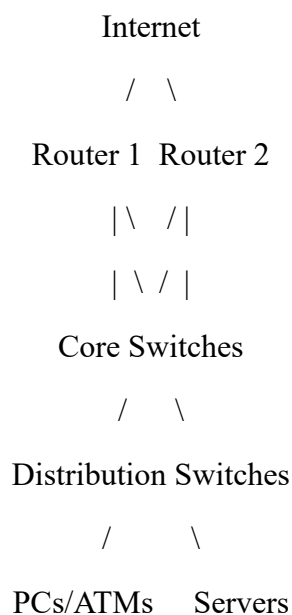
Proposed Solution

A partial mesh topology is recommended for the bank's core network. Core routers, switches, and firewalls are connected through multiple redundant links, providing alternative paths if one link fails.

Network Components

- Enterprise routers
- Core and distribution switches
- Firewalls
- Servers
- Fiber-optic cables
- UPS and redundant power supplies

Network Design



Implementation

1. Create the topology in Cisco Packet Tracer.
2. Connect routers and core switches using redundant links.
3. Connect distribution switches to both core switches.
4. Configure IP addresses and OSPF routing.

5. Test connectivity using ping.
6. Disconnect one link and verify that communication continues through the alternate path.

Problem-Solving Benefits

- Provides high availability.
- Eliminates single points of failure.
- Supports automatic rerouting.
- Provides reliable and high-speed communication.
- Reduces network downtime.

Expected Result

The mesh topology allows banking services to continue even when a network link fails, providing a reliable, fault-tolerant, and highly available network.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Industry Problem

A small classroom computer lab needs a low-cost network for basic activities such as internet browsing and file sharing.

Proposed Solution

A bus topology can be used because all computers share a common communication backbone, reducing hardware and cabling costs.

Network Components

- Coaxial cable (RG-58) or Cat5e/Cat6 cable
- T-connectors and BNC connectors
- 50-ohm terminators for classic coaxial bus networks
- Computers
- Optional low-cost unmanaged switch for a modern simplified design

Network Design

[PC1]---[PC2]---[PC3]---[PC4]---[PC5]

Common Bus Cable

Implementation

1. Connect the computers along a common backbone cable.
2. Use appropriate connectors and terminators for a coaxial bus.
3. Configure IP addresses for all computers.
4. Test communication using the ping command.
5. Use file sharing for resource sharing between computers.

Problem-Solving Benefits

- Low hardware cost
- Simple installation
- Requires less cabling
- Suitable for small classrooms
- Easy resource sharing

Limitation

A break in the main cable can affect the entire network, and network performance decreases as traffic increases.

Expected Result

The bus topology provides a simple and economical network suitable for a small classroom lab with light network usage.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

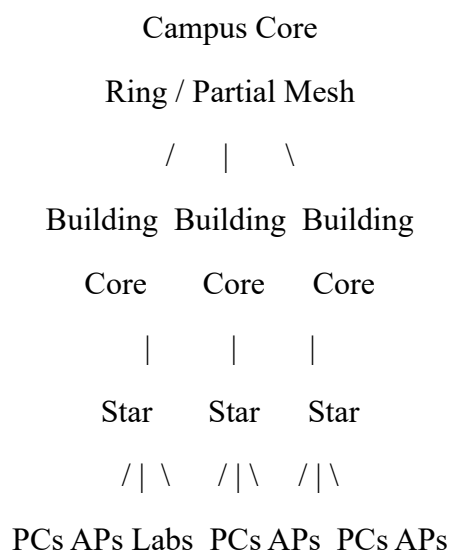
Industry Problem

A large university has thousands of users across lecture halls, laboratories, libraries, hostels, and administrative buildings. A single topology cannot efficiently meet all these requirements.

Proposed Solution

A hybrid topology is proposed by combining mesh/ring, star, and tree structures. This provides reliability, easy management, and scalability.

Network Design



Network Components

- Core routers and multilayer switches
- Distribution switches
- Access switches
- Fiber-optic backbone
- Cat6 UTP cables
- Wireless Access Points
- PoE-enabled switches

Implementation

1. Connect campus buildings to the core using dual fiber-optic links.
2. Use a ring or partial mesh at the campus backbone.
3. Install a distribution switch in each building.
4. Connect floor-level access switches using a star topology.
5. Connect PCs and other devices using Cat6 cables.
6. Connect wireless access points using PoE.
7. Configure IP addressing and routing.
8. Test connectivity using ping.

Problem-Solving Benefits

- High reliability through redundant backbone links.
- Easy management using star topology within buildings.
- Scalability for adding new buildings and devices.
- High-speed communication using fiber optics.
- Wireless connectivity across the campus.
- Fault isolation prevents one building's problem from affecting the entire campus.

Expected Result

The hybrid topology provides a reliable, manageable, and scalable campus network. It combines backbone redundancy with simple building-level networking, making it suitable for a large university.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

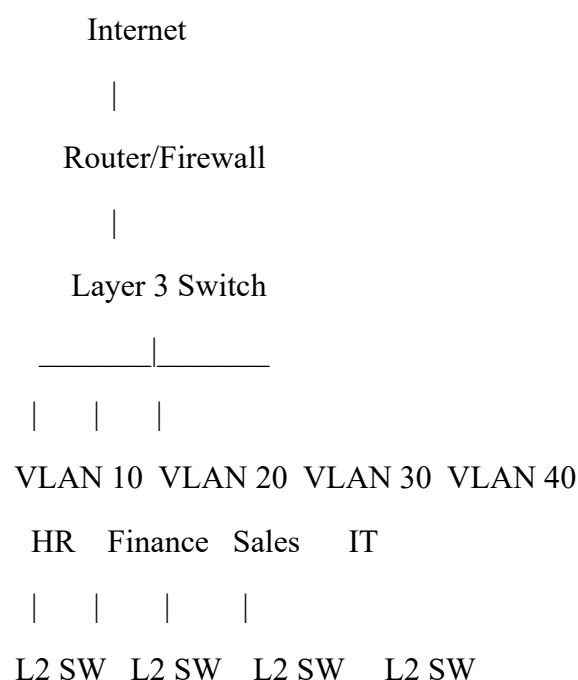
Industry Problem

A company has multiple departments such as HR, Finance, Sales, and IT. The network must provide efficient communication while maintaining security, reliability, and easy management.

Proposed Solution

A hierarchical network architecture using Layer 2 access switches and a Layer 3 core switch/router is proposed. Each department is assigned a separate VLAN to logically isolate its devices.

Network Design



PCs PCs PCs PCs

Implementation

1. Install a Layer 2 switch for each department.
2. Create separate VLANs for HR, Finance, Sales, and IT.
3. Connect the access switches to a Layer 3 switch.
4. Configure **inter-VLAN routing**.
5. Connect the Layer 3 switch to a router/firewall for Internet access.
6. Assign appropriate IP addresses and subnet masks.
7. Test communication using the ping command.

Problem-Solving Benefits

- Improved security through VLAN separation.
- Reduced broadcast traffic.
- Efficient communication between departments.
- Easy troubleshooting and management.
- Scalable when new departments or devices are added.
- Reliable with redundant links and backup paths.

Expected Result

The proposed network provides secure, efficient, reliable, and scalable communication between company departments while keeping departmental traffic logically separated

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and	Good understanding	Basic understanding;	Poor or incorrect

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
		deeply analyzes the industry problem with all factors	with minor gaps	some key aspects missing	understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution